

Master Thesis Coding Context and Requirements

1. Thesis Topic

My master thesis studies the relationship between geopolitical conflict, media expectations, and defense stock prices.

Working title:

Geopolitical Conflicts and Defense Stock Pricing: Real Events vs. Market Expectations

The main objective is to test whether defense and defense-related stocks react more strongly to:

1. **Real conflict intensity**

Measured using daily conflict-event data from ACLED.

2. **Media-driven geopolitical expectations**

Measured using GPR and/or GDELT news/media data.

3. **Potential extension: procurement / military sales announcements**

Measured using Foreign Military Sales notifications or related procurement-event data, if feasible.

The main setting is the **Russia-Ukraine war**, with a baseline focus on Ukraine conflict events from February 2022 onward.

2. Research Question

Main research question:

Do defense and defense-related stocks respond more strongly to realized conflict intensity or to media-driven geopolitical expectations?

Secondary questions:

1. Are responses heterogeneous across different types of defense-related firms?
 2. Do US and European defense-related firms react differently?
 3. Are stock-market reactions driven by battlefield reality, media narratives, or concrete procurement-demand signals?
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3. Contribution

The thesis aims to contribute by separating three channels:

Channel	Data source	Interpretation
Real conflict intensity	ACLED	Actual battlefield escalation
Media expectations	GPR / GDELT	News attention and narratives about future defense demand
Procurement signals	FMS / Bruegel dataset, if feasible	Concrete public signals of military demand

The core novelty is to distinguish **real conflict events** from **media-based expectations** at high frequency and link them to defense stock returns.

4. Data Sources

4.1 Financial Data

Main financial data source:

- Bloomberg

Potential fallback / prototype:

- Yahoo Finance

The financial dataset should be a **firm-day panel**.

Expected period:

- **01/01/2020 to latest available date**

Reason:

- 2020–2021 provides a pre-war baseline.
- February 2022 onward captures the full-scale Russia–Ukraine war.
- This period gives enough daily observations for market-model estimation and panel regressions.

4.2 Firm Universe

The defense-company universe should be identified objectively using Bloomberg index constituents.

Use the following Bloomberg indices:

1. WAERLST Index

Bloomberg World Aerospace & Defense Index

2. BSHIELDT Index

Bloomberg Europe Defense Select Total Return Index

Optional / if available:

- BSHIELDN Index: net return version of the same Europe Defense Select index family.
- MSCI Aerospace & Defense indices were considered as a cross-check, but constituent data may not be available through Bloomberg access.

Baseline firm universe:

Unique listed firms appearing in WAERLST and/or BSHIELDT, subject to available daily stock data.

4.3 Financial Variables to Extract

For each firm:

- Bloomberg ticker
- Company name
- Country
- Exchange
- Currency
- Market capitalization
- BICS sector
- BICS industry
- BICS sub-industry, if available
- Index membership: WAERLST, BSHIELDT, or both
- Index weight, if available

Daily time-series data:

- Adjusted close price or total return price
- Daily trading volume
- Daily market capitalization
- Currency

Market controls:

- S&P 500 index
- STOXX Europe 600 or MSCI Europe
- VIX
- Brent oil price

- EUR/USD exchange rate, optional
-

5. ACLED Data

ACLED is used to measure **real conflict intensity**.

Geographic focus:

- Ukraine

Baseline period:

- February 2022 to latest available date

Raw ACLED fields expected:

- event date
- country
- event type
- sub-event type
- fatalities
- latitude
- longitude
- actor information, if available

Expected aggregation:

Create daily Ukraine-level variables:

- `daily_conflict_events`
- `daily_battles`
- `daily_explosions_remote_violence`
- `daily_shelling_missile_drone_proxy`
- `daily_fatalities`

Main war-intensity index:

$$WI_t = \log(1 + \text{daily_conflict_events})$$

Possible robustness version:

$$WI_severity_t = \log(1 + \text{daily_conflict_events}) + \log(1 + \text{daily_fatalities})$$

Reason for log:

- ACLED event counts can have large spikes.
 - Log transformation reduces the influence of extreme days.
 - +1 allows zero-event days.
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6. GDELT / GPR Data

Media data is used to measure **media-driven geopolitical expectations**.

6.1 GPR

GPR is an existing news-based geopolitical risk index from Caldara and Iacoviello.

Use as:

- baseline media/geopolitical-risk benchmark
- daily series

6.2 GDELT

GDELT is used to build a topic-specific media expectation index.

Possible keyword topics:

- Ukraine war
- NATO rearmament
- military aid
- defense spending
- weapons demand
- missiles
- drones
- air defense
- ammunition
- cybersecurity
- surveillance
- procurement / contracts

Expected GDELT output:

Daily article counts by query/topic.

Main media-expectation index:

```
GEI_t = log(1 + daily_article_count)
```

Possible topic-specific indices:

```
GEI_ukraine_war_t = log(1 + daily_articles_ukraine_war)
GEI_rearmament_t = log(1 + daily_articles_rearmament)
GEI_weapons_demand_t = log(1 + daily_articles_weapons_demand)
GEI_military_aid_t = log(1 + daily_articles_military_aid)
GEI_procurement_t = log(1 + daily_articles_procurement)
```

Reason for log:

- Media article counts can have very large spikes.
- Logs reduce the influence of extreme news days.
- `+1` allows days with zero relevant articles.

7. SIPRI Data

SIPRI can be used as a validation/exposure source.

Most useful SIPRI source:

- SIPRI Arms Industry / Top arms-producing companies

This gives:

- company arms revenue
- arms revenue ranking
- possibly arms revenue share if total revenue is available

Use in the thesis:

1. Validate that Bloomberg-index firms are truly defense-related.
2. Build a firm-level defense exposure measure, if possible.

Preferred exposure measure:

```
DefenseExposure_i = ArmsRevenue_i / TotalRevenue_i
```

Fallback variables:

SIPRI_dummy_i = 1 if firm appears in SIPRI arms-producing companies ranking
SIPRI_arms_revenue_i = company arms revenue
SIPRI_rank_i = company rank by arms revenue

8. Dependent Variables

8.1 Daily Return

From adjusted prices:

$$R_{it} = \log(P_{it}) - \log(P_{i,t-1})$$

Where:

- P_{it} = adjusted price of firm i on day t

8.2 Abnormal Return

The main dependent variable for panel regressions should be:

$$AR_{it}$$

Meaning:

Daily abnormal return of firm i on day t .

Compute abnormal return using a market model:

$$R_{it} = \alpha_i + \beta_i * R_{mt} + \text{error}_{it}$$
$$AR_{it} = R_{it} - \text{predicted}_{R_{it}}$$

Use region-specific benchmarks:

Firm region	Market benchmark
US firms	S&P 500
European firms	STOXX Europe 600 or MSCI Europe

Alternative benchmark:

- MSCI World as robustness.

8.3 CAR for Event Study

For event-study analysis:

$CAR_i[a,b] = \text{sum of } AR_{it} \text{ from event day } a \text{ to event day } b$

Possible windows:

- [-1, +1]
- [-3, +3]
- [-5, +10]

Use CAR only for event-study outputs, not for the main daily panel regression.

9. Explanatory Variables

Main explanatory variables:

WI_t

War intensity from ACLED.

GEI_t

Media expectation index from GDELT/GPR.

DefenseExposure_i

Firm-level defense exposure from SIPRI or Bloomberg/index-based classification.

Main interaction terms:

WI_t * DefenseExposure_i
GEI_t * DefenseExposure_i

Interpretation:

- $WI_t * DefenseExposure_i$ tests whether more defense-exposed firms react more strongly to real battlefield escalation.
- $GEI_t * DefenseExposure_i$ tests whether more defense-exposed firms react more strongly to media expectations.

10. Control Variables

Potential controls:

```
market_return_t
VIX_t
Brent_oil_return_t
log_market_cap_it
log_volume_it
EUR_USD_return_t
```

Interpretation:

- Market return controls for general stock-market movements.
- VIX controls for market uncertainty.
- Brent oil return controls for energy/geopolitical macro shocks.
- Market cap controls for firm size.
- Volume controls for liquidity/investor attention.
- EUR/USD controls for exchange-rate effects, if needed.

11. Main Econometric Model

Baseline panel regression:

```
AR_it = alpha_i + delta_t
        + beta_1 * (WI_t * DefenseExposure_i)
        + beta_2 * (GEI_t * DefenseExposure_i)
        + gamma * X_it
        + epsilon_it
```

Where:

- AR_{it} = abnormal return of firm i on day t
- $alpha_i$ = firm fixed effects

- `delta_t` = date fixed effects or time controls
- `WI_t` = war-intensity index
- `GEI_t` = media-expectation index
- `DefenseExposure_i` = defense exposure of firm i
- `X_it` = control variables
- `epsilon_it` = error term

Cluster standard errors by firm.

Possible robustness:

- Cluster by firm and date.
- Use alternative market benchmarks.
- Use alternative GEI construction.
- Use only SIPRI-confirmed defense firms.
- Split US vs Europe.
- Split high vs low defense exposure.
- Use different event windows.

12. Event Study

Use event study as a complementary method.

Possible event types:

1. Full-scale Russia invasion: 24 February 2022
2. Major escalation days identified from ACLED
3. Major NATO / EU / US defense-policy announcements
4. Major military-aid announcements
5. FMS / procurement notifications, if feasible

Event-study goal:

Estimate whether defense-related firms experience abnormal returns around specific geopolitical or procurement events.

Output:

- CAR by event window
 - average CAR by firm group or defense exposure
 - comparison of high vs low defense-exposure firms
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13. Potential FMS / Procurement Extension

Bruegel's FMS dataset may be used as an extension.

The idea:

FMS notifications are public procurement-related signals. If a company is named as a contractor, investors may expect future revenue.

Possible variable:

$FMS_event_it = 1$ if firm i is named in an FMS notification on date t

Possible event study:

CAR around FMS notification date for named contractors

Comparison group:

- defense firms not named in the notification

This would test whether procurement announcements are priced as concrete military-demand signals.

14. Machine Learning / NLP Extension

ML should be used as a tool, not as the main research question.

Best ML/NLP idea:

Use NLP to classify GDELT news into different narrative topics.

Possible topic categories:

- battlefield escalation
- NATO rearmament
- military aid
- defense spending
- weapons shortages
- procurement/contracts
- drones / AI / cyber / surveillance

Simpler approach:

- keyword dictionaries

More advanced approach:

- topic modeling
- zero-shot classification
- BERTopic
- FinBERT-style sentiment, if appropriate

The goal is to improve the media-expectation index by measuring not only the quantity of media coverage, but also the type of narrative.

15. Expected Data Structure

Final merged dataset should be a firm-day panel:

```
date
ticker
company
country
currency
index_membership
BICS_sector
BICS_industry
price
return
abnormal_return
volume
market_cap
market_return
VIX
Brent_oil_return
WI
GEI
DefenseExposure
WI_x_DefenseExposure
GEI_x_DefenseExposure
```

16. Coding Requirements

Please help me write modular Python code with clear sections:

1. Load Bloomberg firm metadata.
2. Load Bloomberg daily stock prices.
3. Clean tickers and remove duplicates.
4. Compute daily log returns.
5. Load market benchmark data.
6. Estimate market-model expected returns.
7. Compute abnormal returns.
8. Load ACLED conflict data.
9. Aggregate ACLED to daily war-intensity variables.
10. Load GDELT/GPR data.
11. Build media expectation indices.
12. Load SIPRI data, if available.
13. Match SIPRI arms revenue to Bloomberg tickers.
14. Construct defense-exposure variables.
15. Merge all datasets into a firm-day panel.
16. Run baseline panel regressions.
17. Run robustness checks.
18. Run event-study analysis.
19. Export clean datasets and regression outputs.

Preferred output folders:

```
/data/raw  
/data/processed  
/output/tables  
/output/figures  
/scripts
```

Preferred code style:

- use pandas
- use numpy
- use statsmodels or linearmodels for panel regressions
- use clear comments
- avoid hardcoded file paths where possible
- save intermediate cleaned files
- create reproducible outputs
- include checks for missing values and duplicate rows

17. Important Research Logic

The thesis should not simply show that defense stocks go up during war.

The real question is:

Which type of information moves defense stocks more: real battlefield intensity, media narratives, or concrete procurement signals?

The analysis should therefore focus on separating channels:

1. Real conflict channel: ACLED
2. Media expectations channel: GPR / GDELT
3. Procurement channel: FMS / Bruegel, if feasible

The strongest final interpretation would be:

Defense stock prices may be driven not only by actual battlefield escalation, but also by investor expectations about future defense demand.